United States Patent 12404050
Kind Code B2
Date of Patent September 02, 2025
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Device for launching a drone using spring thrust, method for launching drones using this device, and tool for implementing this method

Abstract

A device for launching a drone using spring thrust, in which the means that control the unlocking of an ejection assembly from its armed position include a release housing containing a release chamber, at least one release element which, with the housing, delimits the chamber and is able to move, under the action of the pressure of a gas introduced into the chamber, from a position of rest to a release position in which the at least one release element unlocks the ejection assembly, and return means for returning the at least one release element to the position of rest. The invention also relates to a method for launching drones using the device and to a tool for implementing this method.

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Appl. No.: **18/841793**

Filed (or PCT Filed): **January 09, 2023**

PCT No.: **PCT/IB2023/050148**

PCT Pub. No.: **WO2023/161729**

PCT Pub. Date: August 31, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250162738 A1	May. 22, 2025

Foreign Application Priority Data

FR	2201679	Feb. 28, 2022

Publication Classification

Int. Cl.: **B64U70/70** (20230101)

U.S. Cl.:

CPC **B64U70/70** (20230101);

Field of Classification Search

CPC: B64U (70/70); B64U (70/50); B64F (1/04); B64F (1/06)

USPC: 244/63

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) Not Applicable
STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(3) Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A READ-ONLY OPTICAL DISC, AS A TEXT FILE OR AN XML FILE VIA THE PATENT ELECTRONIC SYSTEM

(4) Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR A JOINT INVENTOR

(5) Not Applicable

BACKGROUND OF THE INVENTION

Field of the Invention

(6) The technical field of the invention is that devices for launching drones, and more specifically, that of devices for launching drones using spring thrust.

(7) Drones, or remotely piloted aircrafts, can carry a payload for civil or military surveillance, intelligence, combat or transport missions. As small flying machines that are cheaper and simpler to operate than aircraft with pilots, drones are enjoying a boom.

Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

(8) A number of devices for launching drones are known, including launch devices with launch tube.

(9) The aim of these devices with launch tube is to propel the drone a few metres above the vehicle carrying the launch tube, for deployment of the drone. This avoids any contact between the drone and the antennae mounted on the vehicle and allows rapid deployment.

(10) There are many types of launch devices with launch tube, including the spring thrust-type device in which a compression spring bears against the rear end of a launch tube and presses against an ejection element slidably mounted in the launch tube and against which the drone will be placed. The device is armed by compressing the compression spring and then locking the ejection element in position, the device then being in the armed position. Unlocking is controlled by a release mechanism, which causes the release of the spring and the launching of the drone, pushed by the ejection element.

(11) This type of launch device is disclosed in Chinese patent application CN111572801 A, in which the release mechanism is a rotary mechanism driven in rotation by a motor. Similarly, the compression of the spring, after the drone has been launched, is implemented by a motor integrated into the device and located in the launch tube.

(12) However, the use of such motorised means increases the cost and weight of the launch device, and does not make it possible to achieve optimum compactness, particularly as it limits the volume that can accommodate a drone in the launch tube.

BRIEF SUMMARY OF THE INVENTION

(13) It is therefore the aim of the invention to provide a launch device that does not have these disadvantages.

(14) The solution according to the present invention is based on the use of a release mechanism comprising at least one release element whose displacement leads to the unlocking of the ejection element and is obtained by the action of a gas pressure in a release chamber, which makes it possible to reduce the cost and weight of the device, and on the use of a separate tool, not integrated into the launch tube, to return the device to the armed position, which makes it possible to increase the volume available for the drone in the launch tube.

(15) The present invention therefore relates to a device for launching a drone using spring thrust, the device comprising a longitudinal launch tube having a first rear end, which is closed, and a second front end, which is open, and the inside of which defines a launch chamber intended to receive a drone, and means for ejecting the drone from the launch tube which comprise: an ejection

assembly comprising an ejection element, mounted so as to slide longitudinally in the launch chamber and intended to push the drone in order to eject it from the launch tube, and a compression spring, referred to as an ejection spring, having a longitudinal axis coaxial with the longitudinal axis of the launch tube and a first end of which bears on the rear end of the launch tube and a second end of which bears against the ejection element, the ejection assembly being able to be placed in an armed position, in which the ejection spring is compressed by the ejection element, locking means for releasably locking the ejection assembly in the armed position, the locking means comprising at least one retaining element connected to the launch tube and movable between a locking position, in which the at least one retaining element is engaged with the ejection element so as to hold the ejection assembly in the armed position, against the action of the ejection spring, and an unlocking position, in which the at least one retaining element is disengaged from the ejection element, and control means for controlling unlocking of the ejection assembly, by moving the at least one retaining element from the locking position to the unlocking position, characterized in that the control means comprise: a release housing, located at the rear of the launch tube and containing a release chamber able to be connected to a gas pressure source, at least one release element which, together with the release housing, delimits the release chamber and is movable, under the action of a gas pressure introduced into the release chamber, from a rest position in which the at least one release element allows the at least one retaining element to remain in the locking position, to a release position, in which the at least one release element moved the at least one retaining element from the locking position to the unlocking position, and means for returning the at least one release element to the rest position.

(16) Said return means may be elastic means, such as a spring.

(17) Preferably, the control means are configured so that the at least one release element moves the at least one retaining element from the locking position to the unlocking position by translational movement of the at least one release element. According to a particular embodiment, the release housing is formed as a hollow cylinder that is integral with the launch tube and has a longitudinal axis, and the release element is a release piston mounted in the release housing so as to slide along said longitudinal axis, the release chamber being cylindrical in shape and delimited radially by the release housing and axially on one side by the release housing and on the other side by the release piston.

(18) However, the present invention is not limited to the way in which the at least one release element moves.

(19) A release element could thus be provided, which is mounted so as to be rotatable relative to the release housing, about the longitudinal axis of the release housing. Such a release element could, for example, comprise a vane-forming part received in the release housing and forming a wall of the release chamber, so that the entry of gas into the release chamber causes the release element to rotate. The return means may, for example, consist in a torsion spring extending around a vertical bearing on which the release element is rotatably mounted, and one arm of which is fixed and the other arm of which is integral with said vane-forming part. Stops may be carried by the release housing so as to define said rest position and an end-of-travel position for the release element. The at least one retaining element will be configured to be moved from the locking position to the unlocking position by rotation of the release element. For example, each retaining member is pivotally mounted in a manner similar to the embodiment to be described below, and the release element will include, extending out of the release housing, for each retaining member, a cam-like member dimensioned to push on the respective retaining element when the release element is rotated.

(20) To further improve the compactness of the ejection means, advantageously the release housing is surrounded by the ejection spring, the at least retaining element is connected to the release housing and is located in the interior space of the ejection spring, and the ejection element comprises a thrust head that is perpendicular to the longitudinal axis of the launch tube and against

which the second end of the ejection spring bears, and at least one holding member that is integral with the thrust head and extends into the interior space of the ejection spring, the at least one retaining element being configured to hold the ejection assembly in the armed position by gripping the at least one holding member in the locking position.

(21) Thus, according to a particular advantageous embodiment of the ejection assembly, in which the release element is a release piston movable in translation, the release housing has a first rear axial side, which is closed, and a second front axial side, which is open and on which is removably mounted a plug comprising a through bore that is coaxial with the longitudinal axis of the launch tube, the release chamber being delimited axially by said first axial side of the release housing and by a first axial side, closed, of the release piston, which is formed as a hollow cylinder, the second front axial side of which is open and formed by an annular end, and the cylindrical wall of which, which defines an interior space, slides against the cylindrical wall of the release housing between the rest position, in which the release piston abuts by its first axial side against a shoulder of the release housing, and the release position, in which the release piston abuts against an axial stop carried by the plug, the at least one retaining element being carried by the plug and, in the armed position, the at least one holding member extending into the release housing through the through bore of the plug so that it can be gripped by the at least one retaining element in the locking position, the at least one retaining element being configured to be located, in the locking position, on the path followed by the annular end of the release piston during its movement from the rest position to the release position, so as to be moved from the locking position to the unlocking position by the annular end of the release piston pushing against the at least one retaining element.

(22) Preferably, the device comprises a plate which is located in the interior space defined by the cylindrical wall of the release piston and is fixed to the plug, and at a distance therefrom, by spacers, the plate thus having a first face facing the first axial side of the release piston and a second opposite face on the side of which the at least one retaining element is located, the axial stop being carried by the first face of the plate and the means for returning the release piston to the rest position being elastic return means, such as a compression spring, bearing on the first face of the plate and pressing against the first axial side of the release piston.

(23) The present invention is also not limited to the way in which the at least one retaining element moves between the locking position and the unlocking position. It could also be contemplated that each retaining element moves in translation.

(24) Preferably, the at least one retaining element is movable between the locking position and the unlocking position by pivoting.

(25) Preferably, the device comprises two retaining elements each pivotally mounted about a respective pivot pin which is integral with said plug, each pivot pin belongs to a plane perpendicular to the longitudinal axis of the launch tube and is perpendicular to a longitudinal plane of the launch tube, the device further comprising means for elastically returning the retaining elements to the locking position.

(26) The holding member can take the form of a rod provided with a circumferential groove able to receive one end of each at least one retaining element in the locking position.

(27) The present invention also relates to a method for launching drones using a launch device using spring thrust as defined above, characterized in that it comprises the following successive steps: a positioning step, comprising positioning a drone in the launch chamber, the ejection assembly being locked in the armed position; a step of launching the drone, comprising introducing into the release chamber sufficient gas pressure to move the release element from the rest position to the release position, whereby the ejection assembly is unlocked and ejects the drone from the launch tube; and before the positioning step is repeated, a repositioning step, comprising returning the ejection assembly to the armed position, in which it is locked by the at least one retaining element.

(28) The present invention also relates to a tool for implementing the method as defined above,

characterized in that it comprises a carriage movable in translation along at least one frame which has, at one end, a drive plate, the carriage and the frame being able to be introduced into the launch chamber, after the launching step, such that the carriage bears against the ejection element and the drive plate bears against the open end of the launch tube, translation driving means being provided for driving the carriage in translation along the frame, by means of which the carriage is movable towards the rear end of the launch tube so as to compress the ejection spring and return the ejection assembly to the armed position.
(29) Preferably, the frame also comprises a bottom plate able, in use, to bear against the rear end of the launch tube, and at least one guide rod connecting the bottom plate and the drive plate and along which the carriage is mounted movable in translation, the translation driving means comprise a worm screw and means for driving the worm screw in rotation, the worm screw extending between the bottom plate and the drive plate, to which it is connected so as to be rotatable about its longitudinal axis, and passing through a threaded hole of the carriage, whereby rotation of the worm screw results in translation of the carriage.
(30) The means for driving in rotation can be manual or motorised.
(31) The means for driving in rotation may comprise a gear set mounted in the drive plate and connected to a drive shaft accessible from outside, for manual drive, for example by a crank, or motorised drive, for example using a drill.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) To better illustrate the subject-matter of the present invention, a particular embodiment will be described below, with reference to the attached drawings. In these drawings:
(2) FIG. **1** is a side view, in longitudinal section, of the launch device according to the particular embodiment of the present invention, in the armed position;
(3) FIG. **2** is a side view, in longitudinal section, showing more specifically the ejection means of the device in FIG. **1**, in a sectional plane perpendicular to the sectional plane in FIG. **1**;
(4) FIG. **3** is a perspective view of the locking means for locking in the armed position;
(5) FIG. **4** is an exploded perspective view of the locking means in FIG. **3**;
(6) FIG. **5** is a perspective view of the tool according to the particular embodiment of the present invention;
(7) FIG. **6** is a perspective view of the tool in FIG. **5**, mounted in the launch chamber of the device according to the invention, with the launch tube omitted; and
(8) FIG. **7** is a side view, in longitudinal section, of the tool and the launch device in the same position as shown in FIG. **6**.
DETAILED DESCRIPTION OF THE INVENTION
(9) Referring first of all to FIG. **1**, it can be seen that the launch device **1** according to the present invention is intended for launching a drone-type object **2**.
(10) Such a drone **2** typically comprises a base body **20** containing a power plant, a battery pack and navigation electronics. The drone **2** is equipped with a payload **21** removably mounted on the base body **20**. This payload **21** may be a lethal or non-lethal load. For example, the drone **2** could be equipped with an explosive-type lethal load, a non-lethal load enabling paint or smoke to be released, or an optronic load enabling observation and detection.
(11) Referring now to FIGS. **1** to **4**, it can be seen that the device **1** comprises a launch tube **3** with a longitudinal axis A**1**, intended to receive the object to be launched, i.e. the drone **2**, before it is launched, and ejection means **4** for ejecting the drone **2**.
(12) The launch tube **3** has a generally cylindrical shape about its longitudinal axis A**1**, and it has a front end **3***a* forming the mouth for the exit of the drone **2** and a rear end **3***b* in the region of which

the ejection means **4** are arranged.
(13) It is emphasized here that the terms “front” and “rear” refer to the predetermined direction of movement of the drone **2** relative to the launch tube **3** during launch.
(14) The front end **3***a* is closed, prior to launch, by a circular tight plug **30**, the diameter of which corresponds to the exterior diameter of the launch tube **3**.
(15) The rear end **3***b* is closed by a base **31** in the form of a block of circular section, the diameter of which corresponds to the external diameter of the launch tube **3**. The base **31** is fixed to an interior shoulder **32** of the launch tube **3**, which is annular in shape, by fasteners **33**, such as screws, passing through orifices provided in the base **31** and orifices provided in the interior shoulder **32** and located opposite each other.
(16) The launch tube **3** defines therein a cylindrical launch chamber **34** having, in the region of the rear end **3***b*, a seat **35**, here formed by the interior shoulder **32**, against which the drone **2** will be placed before ejection from the launch tube **3** by the ejection means **4**.
(17) The ejection means **4** comprise an ejection assembly **5** able to be placed in an armed position, locking means **6** for releasably locking the ejection assembly **5** in the armed position, and control means **7** for controlling the unlocking of the ejection assembly **5**.
(18) As it can be seen in FIGS. **1** and **2**, the ejection assembly **5** comprises an ejection element **50** and an ejection spring **51** which consists of a compression spring.
(19) In this particular embodiment, the ejection element **50** is formed as an ejection piston **52** having a piston body **53**, a thrust head **54** and a holding rod **55**.
(20) The piston body **53** is a hollow cylindrical body having a first axial side **53***a*, directed towards the rear end **3***b* of the launch tube **3**, which is open, and a second axial side, directed towards the front end **3***a* of the launch tube **3**, which is closed by a circular transverse wall **53***b* having, at its centre, an orifice. An annular central projection **53***c* surrounding the orifice extends inside the ejection piston **52**, from the transverse wall **53***b*. The transverse wall **53***b* also carries an annular exterior projection **53***d* arranged and dimensioned so that the ejection spring **51** is received between the exterior projection **53***d* and the internal side wall of the piston body **53**.
(21) The thrust head **54** is a hemispherical body integral with the transverse wall **53***b*, on the exterior side of the piston body **53**. The diameter of the base of this hemispherical body is equal to the exterior diameter of the piston body **53**. The thrust head **54** has, at its centre, an orifice **54***a* located opposite the orifice of the transverse wall **53***b* and dimensioned to receive and lock the holding rod **55**, which is thus made integral with the ejection piston **52**.
(22) The holding rod **55** extends along a longitudinal axis A**2** coaxial with the longitudinal axis A**1** and slidably extends into a cylindrical bore **56** formed in a plug **57**.
(23) This plug **57** extends transversely to the launch chamber **35**, inside the piston body **53**. The plug **57** is fixed with respect to the launch tube **3**. In particular, the plug **57** is fixed, with radial screws **58**, to an open front side of a tubular housing **37**, or case, extending from the base **31**.
(24) The housing **37** extends perpendicularly from the transverse wall of the base **31**, in other words parallel to said longitudinal axis A**1**, towards the front end **3***a* of the launch tube **3**. Preferably, the housing **37** and the base **31** are one piece.
(25) The housing **37** is dimensioned so that it is contained within the piston body **53** in the armed position of the ejection assembly **5**.
(26) The holding rod **55** has, at its free end region, opposite its end region connected to the piston body **53**, a groove **55***a* intended to receive the locking means **6** in the locking position. The groove **55***a* is formed over the entire circumference of the holding rod **55**.
(27) The ejection spring **51** is located in the space formed between the exterior wall of the housing **37** and the interior wall of the piston body **53**, between the rear end **3***b* of the launch tube **3** and the transverse wall **53***b* of the piston body **53**. More specifically, the ejection spring **51** has a first end bearing on the transverse wall of the base **31** and a second end bearing against the transverse wall **53***b*, made integral with them for example by any suitable means.

(28) The locking means **6** keep the ejection spring **51** in the compressed state and therefore prevent the ejection piston **52** from sliding towards the front end of the launch tube **3**, thereby keeping the ejection assembly **5** in the armed position.
(29) In the particular embodiment shown in FIGS. **1** to **4**, the locking means **6** comprise two retaining elements **60**.
(30) Each retaining element **60** is rotatably mounted about a pivot pin **61** integral with the plug **57**. The two pivot pins **61** are parallel to each other. Each pivot pin **61** is integral with two facing plates **59** extending perpendicularly from the transverse wall of the plug **57**. The pivot pins **61** are therefore orthogonal to the longitudinal axis A**1**.
(31) Each retaining element **60** is formed as a finger having a bent section **60***a* and a nose section **60***b*. The pivot pin **61** is received in a through hole **60***c* formed at the corner of the bent section **60***a*. The free end of the bent section **60***a* has a rounded profile. The nose section **60***b* has a substantially V-shaped profile, Each retaining element **60** is dimensioned and arranged such that the free end of the nose section **60***b* is able to be received in the groove **55***a* of the holding rod **55**, and the free end region of the bent section **60***a* is able to come into contact with the interior wall of the housing **37**.
(32) The two retaining elements **60** are arranged in the same plane passing through the longitudinal axis A**1** and perpendicular to the transverse wall of the plug **57**, the free ends of their nose section **60***b* being directed towards each other. Two return springs **62** extend between the two retaining elements **60**. In particular, two pins **60***d* extend on either side of each retaining element **60** at the corner of the nose section **60***b* and parallel to the pivot pins **61**. A return spring **62** is mounted between each facing pin **60***d*. These return springs **62** tend to return the nose sections **60***b* of the retaining elements **60** towards the groove **55***a* of the holding rod **55**, thus urging the retaining elements **60** towards their locking position.
(33) The control means **7** are intended to control the movement of the retaining elements **60** to their unlocking position, against the return springs **62**, by the action of a gas pressure.
(34) The control means **7** comprise a release element **70** and a release chamber **71** able to be connected to a gas pressure source **72**.
(35) The release element **70** is formed as a release piston **73** slidably mounted in the housing **37**. As it can be seen in FIGS. **3** and **4**, the release piston **73** is a hollow cylindrical body having a first axial side, directed towards the rear end **3***b* of the launch tube **3**, which is closed by a circular transverse wall **73***a*, and a second axial side, directed towards the front end **3***a* of the launch tube **3**, which is open and formed by an annular end **73***c*.
(36) The housing **37** has a shoulder **37***a* which defines a first axial stop for the release piston **73**, against which the transverse wall **73***a* is in contact when the release piston **73** is in the rest position.
(37) At its centre, the transverse wall **73***a* has a projecting tubular part forming a counter-stop **73***c* able to come into contact with an axial stop **57***a* integral with the plug **57** in order to limit the travel of the release piston **73**, i.e. its sliding towards the front end of the housing **37**. The axial stop **57***a* is integral with a transverse circular plate **57***b* which is in turn integral, via two spacers **57***c*, with the transverse wall of the plug **57**. The two spacers **57***c* extend on the outside of the plates **59** carrying the pivot pins **61** of the retaining elements **60** and their length is such that the retaining elements **60** cannot come into contact with the plate **57***b*.
(38) A compression spring **74** extends around the axial stop **57***a* and the counter-stop **73***c*, between the plate **57***b* and the transverse wall **73***a* of the release piston **73**. This spring **74** is intended to urge the release piston **73** towards the rear end **3***b* of the launch tube **3**, in other words away from the axial stop **57***a*. Thus, once the release chamber **71** is no longer pressurized, the release piston **73** is automatically returned to its rest position, i.e. to a position in which the release piston **73** is not in contact with the retaining elements **60**.
(39) The release chamber **71** is radially delimited by the interior wall of the housing **37**. The release chamber **71** has a rear end closed by the transverse wall of the base **31** and a front end delimited by the transverse wall **73***a* of the release piston **73**. The base **31** has an inlet orifice opening into the

release chamber **71** and is intended to be connected to a gas pressure source **72** by a pneumatic connector **72***a*.

(40) The gas pressure source **72** is intended to enable the release chamber **71** to be pressurized by compressed gas, which may be air. The pressure source **72** can be any gas pressure source. For reasons of space, the pressure source **72** is an external pressure source, i.e. it is arranged outside the launch tube **3** and is able to be brought into communication with the release chamber **71** via the pneumatic connector **72***a*.

(41) Referring again to FIG. **1**, it can be seen that the device **1** also includes a launch sabot **8** intended to receive the drone **2**.

(42) The launch sabot **8** is dimensioned and configured to be received in the launch chamber **34** to enable the drone **2** to be guided in the launch tube **3** during the launch phase, and to surround the drone **2** to enable the drone **2** to be protected during launch. It should be noted that a different sabot **8** is defined according to the profile of the drone **2** and its payload **21**.

(43) In the particular embodiment shown, the sabot **8** is formed of several separable segments **80**, for example four identical segments **80**. These segments **80** delimit between them a space for receiving the drone **2** having an opening intended to be located opposite the front end **3***a* of the launch tube **3**. These segments **80** also delimit between them a space for receiving the ejection means **4**, having an opening intended to allow the passage of the ejection means **4** when the sabot **8** is introduced into the launch chamber **34**.

(44) Alternatively, the sabot can be formed of four separable segments, three of which are identical and one of which contains a charging and communication device between the drone and the launch tube. As the sabot is designed to be ejected from the launch tube and to disengage from the drone after launch, the charging and communication device is a wireless device. Preferably, this device is of inductive type and a transmission coil B (FIG. **2**) is integral with the base **31** and able to cooperate, via said device, with a receiver coil housed in the drone.

(45) The launch device **1** according to the present invention enables drones **2** to be launched easily, quickly and reliably, the launching method comprising a pre-launch phase, a launch phase and a post-launch phase.

(46) During the pre-launch phase, the launch sabot **8**, in the reception space of which the drone **2** is received, is first inserted through the open front end **3***a* of the launch tube **3**, until it comes into contact with the seat **35**, then the front end **3***a* of the launch tube **3** is closed by the tight plug **30**.

(47) During this phase, the ejection means **4** are placed in the space of the sabot **8** for receiving the ejection means. In particular, the ejection assembly **5** is in the armed position, the ejection spring **51** being compressed, the retaining elements **60** being in the locking position and the release piston **73** being not in contact with the retaining elements **60**.

(48) When it is desired to launch the drone **2**, during the launch phase, the release chamber **71** should simply be pressurized. The pressure contained in the release chamber **71** will then push the release piston **73** towards the retaining elements **60**, until the counter-stop **73***c* of the release piston **73** abuts against the axial stop **57***a*. During this sliding movement, the annular end **73***b* of the release piston **73** comes into contact with the rounded profile end of the retaining elements **60** and pushes them inwards, causing the retaining elements **60** to rotate in a direction tending to move the nose sections **60***b* away from the groove **54***a*. Once the retaining elements **60** are disengaged from the holding rod **55**, i.e. in the unlocking position, the holding rod **55** is released and the ejection piston **52** is free to slide and no longer holds the ejection spring **51** in the compressed state. The ejection spring **51** is therefore released instantaneously and its extension causes the ejection piston **52** to slide suddenly towards the front end **3***a* of the launch tube **3**. In this position, the thrust head **53** applies a thrust force against the sabot **8**, thus ejecting the sabot **8** and the drone **2** contained therein from the launch tube **3**.

(49) Once ejected from the launch tube **3**, the sabot **8** disengages itself into separate segments **80**. Once at its apogee, the drone **2** starts up and stabilizes and its mission can begin.

(50) As it can be seen in FIG. **7**, when the sabot **8** and the drone **2** are ejected from the launch tube **3**, the ejection piston **50** is close to the front end **3***a* of the launch tube **3**, with the ejection spring **51** in its fully extended state. Following the exhausting of the release chamber **71**, the release piston **73** was returned to the rest position by the spring **73**, allowing the return springs **62** to pivot the retaining elements **60** inwards.
(51) During the post-launch phase, the launch device **1** is prepared for the launch of another drone **2**, in other words, the ejection assembly **5** is repositioned to its armed position, during which the ejection spring **51** is compressed again.
(52) This phase is carried out by the tool **9** according to a particular embodiment of the present invention, shown in FIGS. **5** to **7**.
(53) As it can be seen in FIG. **5**, the tool **9** comprises a carriage **10** and a frame **11** along which the carriage **10** is movable in translation by translation driving means.
(54) The translation driving means comprise a worm screw **12** and a gear set **13**.
(55) The worm screw **12** is mounted between a bottom plate **14** and a drive plate **15**, which are connected by two guide rods **16** that form part of the frame **11**.
(56) The bottom plate **14** is a circular plate the external diameter of which is smaller than the internal diameter of the launch tube **3**. The bottom plate **14** has a central cylindrical bore **14***a*, the diameter of which is greater than the exterior diameter of the ejection piston **52**.
(57) The drive plate **15** is also a circular plate able to fit over the open front end **3***a* of the launch tube **3**, and has therefore the same size as the tight plug **30**.
(58) The drive plate **15** incorporates the gear set **13**, one gear of which engages with the thread of the worm screw **12**, so that the worm screw **12** can be rotated by controlling the gear set **13**.
(59) The carriage **10** is traversed by the two guide rods **16**, passing through two bores of the carriage **10**, and by the worm screw **12**, passing through a threaded hole, so that a rotation of the worm screw **12** causes the carriage **10** to move in translation along the guide rods **16**.
(60) The carriage **10** has an exterior diameter equal to the diameter of the bottom plate **14**. The carriage **10** has a receiving part, able to receive and enclose the thrust head **54** and the front end region of the piston body **53**.
(61) In practice, as it can be seen in FIG. **7**, during the post-launch phase, the tool **9** is positioned in the launch tube **3**. More specifically, the bottom plate **14** is positioned against the seat **35**, with its bore **14***a* surrounding the ejection spring **51**, the drive plate **15** is positioned at the front end **3***a* of the launch tube **3** and the carriage **10** covers the thrust head **54**. The gear set **13** is activated, either manually or by motor, to rotate the worm screw **12**. The carriage **10** is then moved in translation towards the rear end **3***b* of the launch tube **3**. As the carriage **10** covers the thrust head **54**, the movement of the carriage **10** causes the thrust head **54** and therefore the holding rod **55** to move progressively towards the rear end **3***b*, thereby compressing the ejection spring **51**. When the holding rod **55** comes into contact with the retaining elements **60**, the holding rod **55** first rotates them against the return springs **62**, then the retaining elements **60** automatically engage in the groove **55***a* as soon as the nose sections **60***b* are facing it. The ejection spring **51** is then placed in its compressed state and the ejection assembly **5** is in its armed position, ready for a new launch phase.
(62) The tool **9** is removed from the launch tube **3** and a new sabot **8** can then be positioned in the launch chamber **35**, and the previous phases are repeated for the launch of this drone **2** and other drones.
(63) It is understood that the particular embodiment described above is indicative and non-limiting, and that modifications may be made without departing from the scope of the present invention.

Claims

1. A device for launching a drone using spring thrust, the device comprising: a longitudinal launch tube having a first rear end, which is closed, and a second front end, which is open, and the inside of the launch tube defining a launch chamber intended to receive the drone, and ejection means for ejecting the drone from the launch tube which the ejection means comprising: an ejection assembly comprising an ejection element, mounted so as to slide longitudinally in the launch chamber and intended to push the drone in order to eject the drone from the launch tube, and an ejection spring having a longitudinal axis coaxial with a longitudinal axis of the launch tube, wherein a first end of the ejection spring bears on the rear end of the launch tube and a second end of the ejection spring bears against the ejection element, the ejection assembly being able to be placed in an armed position, in which the ejection spring is compressed by the ejection element, locking means for releasably locking the ejection assembly in the armed position, the locking means comprising at least one retaining element connected to the launch tube and movable between a locking position, in which the at least one retaining element is engaged with the ejection element so as to hold the ejection assembly in the armed position, against the action of the ejection spring, and an unlocking position, in which the at least one retaining element is disengaged from the ejection element, and control means for controlling unlocking of the ejection assembly, by moving the at least one retaining element from the locking position to the unlocking position, wherein the control means comprise: a release housing, located at a rear of the launch tube and containing a release chamber able to be connected to a gas pressure source, at least one release element which, together with the release housing, delimits the release chamber and is movable, under the action of a gas pressure introduced into the release chamber, from a rest position in which the at least one release element allows the at least one retaining element to remain in the locking position, to a release position, in which the at least one release element moves the at least one retaining element from the locking position to the unlocking position, and means for returning the at least one release element to the rest position.
2. The device according to claim 1, wherein the release housing is formed as a hollow cylinder that is integral with the launch tube and has a longitudinal axis, and the release element is a release piston mounted in the release housing so as to slide along the longitudinal axis of the release housing, the release chamber being cylindrical in shape and delimited radially by the release housing and the release chamber being delimited axially one side by the release housing and another side by the release piston.
3. The device according to claim 2, wherein the release housing is surrounded by the ejection spring, the at least one retaining element is connected to the release housing and is located in an interior space of the ejection spring, and the ejection element comprises a thrust head that is perpendicular to the longitudinal axis of the launch tube and the second end of the ejection spring bears against the thrust head, and at least one holding member that is integral with the thrust head and extends into the interior space of the ejection spring, the at least one retaining element being configured to hold the ejection assembly in the armed position by gripping the at least one holding member in the locking position.
4. The device according to claim 3, wherein the release housing has a first rear axial side, which is closed, and a second front axial side, which is open and on which is removably mounted a plug, the plug comprising a through bore that is coaxial with the longitudinal axis of the launch tube, the release chamber being delimited axially by the first axial side of the release housing and by a first axial side of the release piston, the first axial side of the release piston being closed, the release piston being formed as a hollow cylinder, wherein a second front axial side of the release piston is open and formed by an annular end, and a cylindrical wall of the release piston, which defines an interior space, slides against a cylindrical wall of the release housing between the rest position, in which the first axial side of the release piston abuts against a shoulder of the release housing, and the release position, in which the release piston abuts against an axial stop carried by the plug, the

at least one retaining element being carried by the plug and, in the armed position, the at least one holding member extending into the release housing through the through bore of the plug so that the at least one holding member can be gripped by the at least one retaining element in the locking position, the at least one retaining element being configured to be located, in the locking position, on a path followed by the annular end of the release piston during a movement of the release piston from the rest position to the release position, so as to be moved from the locking position to the unlocking position by the annular end of the release piston pushing against the at least one retaining element.

5. The device according to claim 4, wherein the device comprises a plate which is located in an interior space defined by the cylindrical wall of the release piston, wherein the plate is fixed to the plug, and at a distance therefrom, by spacers, the plate having a first face facing the first axial side of the release piston and a second opposite face on a side of which the at least one retaining element is located, the axial stop being carried by the first face of the plate and the means for returning the release piston to the rest position being elastic return means bearing on the first face of the plate and pressing against the first axial side of the release piston.

6. The device according to claim 4, wherein the at least one retaining element is movable between the locking position and the unlocking position by pivoting.

7. The device according to claim 6, wherein the at least one retaining element includes two retaining elements each pivotally mounted about a respective pivot pin, said pivot pin is integral with the plug, each pivot pin is orthogonal to the longitudinal axis of the launch tube and is perpendicular to a longitudinal plane of the launch tube, the device further comprising means for elastically returning the retaining elements to the locking position.

8. The device according to claim 1, wherein the release housing is surrounded by the ejection spring, the at least one retaining element is connected to the release housing and is located in an interior space of the ejection spring, and the ejection element comprises a thrust head that is perpendicular to the longitudinal axis of the launch tube and the second end of the ejection spring bears against the thrust head, and at least one holding member that is integral with the thrust head and extends into the interior space of the ejection spring, the at least one retaining element being configured to hold the ejection assembly in the armed position by gripping the at least one holding member in the locking position.

9. The device according to claim 1, wherein the at least one retaining element is movable between the locking position and the unlocking position by pivoting.

10. A method for launching drones using a device as defined in claim 1, wherein the method comprises the following successive steps: a positioning step, comprising positioning a drone in the launch chamber, the ejection assembly being locked in the armed position; a step of launching the drone, comprising introducing into the release chamber sufficient gas pressure to move the release element from the rest position to the release position, whereby the ejection assembly is unlocked and ejects the drone from the launch tube; and before the positioning step is repeated, a repositioning step, comprising returning the ejection assembly to the armed position, in which the ejection assembly is locked by the at least one retaining element.

11. A tool for implementing the method as defined in claim 10 for launching drones using the device, wherein the tool comprises: a carriage movable in translation along at least one frame, wherein the frame has, at one end, a drive plate, the carriage and the frame configured to be introduced into the launch chamber such that the carriage bears against the ejection element and the drive plate bears against the front end of the launch tube, and translation driving means being provided for driving the carriage in translation along the frame, the carriage is movable towards the rear end of the launch tube by means of said translation driving means so as to compress the ejection spring and return the ejection assembly to the armed position.

12. The tool according to claim 11, wherein the frame further comprises a bottom plate able to bear against the rear end of the launch tube, and at least one guide rod connecting the bottom plate and

the drive plate, the carriage is mounted movable in translation along the at least one guide rod, the translation driving means comprise a worm screw and means for driving the worm screw in rotation, the worm screw extending between the bottom plate and the drive plate, the worm screw is connected to the bottom plate and to the drive plate so as to be rotatable about a longitudinal axis of the worm screw, and passing through a threaded hole in the carriage, whereby rotation of the worm screw results in translation of the carriage.